

Matching-count sequence of the $4 \times n$ grid graph
with two corner vertices removed: interleave
identity and parity/mod-4 laws
(Lean 4 machine-checked claim of record)

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Abstract

Let $a(n)$ be the number of matchings (the empty one included) of the graph obtained from the $4 \times n$ grid by deleting the upper and the lower corner vertex of the last column; exact enumeration starts 2, 15, 209, 2426, 29566, $\dots$, and the sequence satisfies an order-9 integer recurrence whose two position-parity subsequences satisfy the *same* order-9 recurrence with their own initial values. We state and machine-check, in Lean 4 (v4.34.0, mathlib v4.34.0), three theorems about these recurrence-defined sequences: the main sequence is exactly the interleaving of the two subsequences; its parity depends only on the index modulo 5 (odd if and only if the index is 2 or 3 modulo 5); and its residue modulo 4 depends only on the index modulo 10, with the ten-term pattern (2, 3, 1, 2, 2, 0, 1, 1, 0, 0). Every proof compiles with no **sorry** and the axiom audit reports only {propext, Quot.sound, Classical.choice}. Novelty adjudication: no occupying record found in three consistent rounds of the pipeline’s exclusivity review; value tier *new sequence* only — the order-9 recurrence is the namesake recurrence of OEIS A033507 and the perfect-matching subcolumn is OEIS A129113, so no novelty is claimed for either. The combinatorial identification is not kernel-proved and is stated as a conjecture. This note is a priority claim of record, not a full paper.

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1 Introduction

Three theorems about three integer sequences defined by initial values plus linear recurrences are reported here; the sequences come from counting matchings of the $4 \times n$ grid graph with the upper and lower corner vertices of the last column removed, and the order-9 recurrence was obtained by exact rational minimal-order fitting of enumerated data (the Berlekamp–Massey family of methods [6]); the matching-count context goes back to Heilmann and Lieb and to Godsil and Gutman [4, 5]. The proofs were produced by the AI4Math pipeline, in which LLM workers generate statements and proof scripts and the Lean 4 kernel is the sole adjudicator at every gate: statements are frozen by hash before proving, and a separate audit re-computes the symmetric difference, scans the whole file for `sorry/admit` including comments, and re-prints the axiom dependencies [1, 2]. This note is a priority claim of record deposited so that a dated, citable public record exists ahead of any narrative paper; the fuller paper for the same three theorems is prepared separately in the source repository under `papers/grid4n-col-2notch/` and the two lanes are linked through the deposit DOI. No literature review is attempted here.

2 Statement

In words, the counted object is the 4-by- n grid of squares with the top and bottom corner vertices of the last column removed: one counts the ways to choose some of the edges so that no two chosen edges share an endpoint, with choosing none counted as one way. The count begins 2, 15, 209, 2426, 29566, That this sequence is the matching count of that graph is supported by exact enumeration and is stated here only as Conjecture 1; the frozen statement layer contains no graph, no grid and no matching.

Throughout, the three sequences are *defined* by the displayed initial values and recurrences. The formal layer is 0-indexed, with the index convention carried in the frozen definitions’ header comments: $\mathbf{atc}(n) = a(n+1)$ for the main sequence, $\mathbf{ot}(k) = a(2k+1)$ for the odd-position subsequence and $\mathbf{et}(k) = a(2k+2)$ for the even-position subsequence. The main sequence satisfies the order-9 recurrence

$$\begin{aligned} a(n) = & 9a(n-1) + 41a(n-2) - 41a(n-3) - 111a(n-4) + 91a(n-5) \\ & + 29a(n-6) - 23a(n-7) - a(n-8) + a(n-9) \quad (n \geq 10), \end{aligned} \tag{1}$$

with initial values $a(1..9) = 2, 15, 209, 2426, 29566, 355504, 4290501, 51728089, 623832332$; each position subsequence satisfies the *same* order-9 recurrence

$$b(k) = 163b(k-1) - 2641b(k-2) + 12479b(k-3) - 22577b(k-4) + 16705b(k-5) - 5331b(k-6) + 769b(k-7) - 47b(k-8) + b(k-9) \quad (2)$$

for $b \in \{o, e\}$, with initial values the odd- and even-positioned terms of a . Structurally, the subsequence recurrence is what the main recurrence becomes on the square-conjugate face: it is annihilated by $p(x)p(-x)$, where p is the characteristic polynomial of (1), which has the same order 9 — in contrast to order-dropping interleavings, where the subsequence recurrence would have strictly smaller order. That the subsequence recurrence used here is exactly this pullback is the content of Theorem 1.

Listings 1 and 2 show the three sequence definitions verbatim.

Listing 1: The main sequence `atc`: nine initial values and the order-9 recurrence of (1), verbatim from the frozen statement file `formalized/01-comb09-4ntwocorner-statements.lean` (lines 30–47, sha256 4cb10838). The docstring carries the 0-based convention $\text{atc}(n) = a(n+1)$.

```

1  /-- **序列 atc (主列, 九阶全滞后递推) **.
2  初值  $a(0..8) = 2, 15, 209, 2426, 29566, 355504, 4290501, 51728089,$ 
    $623832332$ 
3  (0-based; 1-based 即  $a(1..9)$ , 逐位取自卡面与 counts.txt) :
4  递推  $\text{atc}(n+9) = 9 \cdot \text{atc}(n+8) + 41 \cdot \text{atc}(n+7) - 41 \cdot \text{atc}(n+6) - 111 \cdot \text{atc}(n$ 
    $+5)$ 
5   $+ 91 \cdot \text{atc}(n+4) + 29 \cdot \text{atc}(n+3) - 23 \cdot \text{atc}(n+2) - \text{atc}(n+1) + \text{atc}(n)$  ( $n$ 
    $\geq 0$ ) 。 -/
6  def atc : ℕ → ℤ
7    | 0 => 2
8    | 1 => 15
9    | 2 => 209
10   | 3 => 2426
11   | 4 => 29566
12   | 5 => 355504
13   | 6 => 4290501
14   | 7 => 51728089
15   | 8 => 623832332
16   | n + 9 =>
17     9 * atc (n + 8) + 41 * atc (n + 7) - 41 * atc (n + 6) - 111
     * atc (n + 5)

```

```

18      + 91 * atc (n + 4) + 29 * atc (n + 3) - 23 * atc (n + 2) -
      atc (n + 1) + atc n

```

Listing 2: The two position subsequences `ot` (odd side) and `et` (even side), verbatim from the frozen statement file (lines 49–87, sha256 4cb10838). Both satisfy the same order-9 recurrence (2); their initial values are the odd- and even-positioned terms of `atc`.

```

1  /-- **序列 ot (奇位侧子列, 九阶递推) **: ot k = a(2k+1)。
2  初值 o(0..8) = 2, 209, 29566, 4290501, 623832332, 90717725366,
3      13192325177001, 1918451920976894, 278984786074420317
4  (= atc 在下标 0,2,...,16 处之值, 即 1-based a(1),a(3),...,a(17)) :
5  递推 ot(k+9) = 163·ot(k+8) - 2641·ot(k+7) + 12479·ot(k+6) - 22577·
      ot(k+5)
6      + 16705·ot(k+4) - 5331·ot(k+3) + 769·ot(k+2) - 47·ot(k+1) + ot(k)
      (k ≥ 0)。 -/
7  def ot : ℕ → ℤ
8      | 0 => 2
9      | 1 => 209
10     | 2 => 29566
11     | 3 => 4290501
12     | 4 => 623832332
13     | 5 => 90717725366
14     | 6 => 13192325177001
15     | 7 => 1918451920976894
16     | 8 => 278984786074420317
17     | k + 9 =>
18         163 * ot (k + 8) - 2641 * ot (k + 7) + 12479 * ot (k + 6) -
19         22577 * ot (k + 5)
20         + 16705 * ot (k + 4) - 5331 * ot (k + 3) + 769 * ot (k +
21         2) - 47 * ot (k + 1) + ot k
22
23 /-- **序列 et (偶位侧子列, 九阶递推) **: et k = a(2k+2)。
24 初值 e(0..8) = 15, 2426, 355504, 51728089, 7522727192,
25     1093972462647,
26     159087517952550, 23134798325751388, 3364304696158749633
27 (= atc 在下标 1,3,...,17 处之值, 即 1-based a(2),a(4),...,a(18)) :
28 递推 et(k+9) = 163·et(k+8) - 2641·et(k+7) + 12479·et(k+6) - 22577·
      et(k+5)
29     + 16705·et(k+4) - 5331·et(k+3) + 769·et(k+2) - 47·et(k+1) + et(k)
      (k ≥ 0)。 -/
30 def et : ℕ → ℤ
31     | 0 => 15
32     | 1 => 2426

```

```

30 | 2 => 355504
31 | 3 => 51728089
32 | 4 => 7522727192
33 | 5 => 1093972462647
34 | 6 => 159087517952550
35 | 7 => 23134798325751388
36 | 8 => 3364304696158749633
37 | k + 9 =>
38   163 * et (k + 8) - 2641 * et (k + 7) + 12479 * et (k + 6) -
   22577 * et (k + 5)
39   + 16705 * et (k + 4) - 5331 * et (k + 3) + 769 * et (k +
   2) - 47 * et (k + 1) + et k

```

Theorem 1 (interleaving). *For every $k \in \mathbb{N}$, the main sequence splits by position parity:*

$$atc(2k) = ot(k) \quad \text{and} \quad atc(2k+1) = et(k).$$

In the 1-based counting convention: $a(2k+1) = ot(k)$ and $a(2k+2) = et(k)$.

Listing 3: Formal statement of Theorem 1, verbatim from the frozen statement file `formalized/01-comb09-4ntwocorner-statements.lean` (lines 89–93, sha256 4cb10838); the `:=` by opens the proof body, which is given in Listing 7.

```

1  /-- **T1 (主攻) 交织恒等式**: 完整九阶递推 = 两条同签名九阶子列的交织
   -----
2  对一切  $k$ ,  $atc(2k) = ot\ k$  且  $atc(2k+1) = et\ k$ 
3  (子列递推 = 主递推特征多项式的平方合取  $p(x)p-(x)$  面, 无阶下降)。
   -/
4  theorem atc_interleave (k : ℕ) :
5    atc (2 * k) = ot k ∧ atc (2 * k + 1) = et k := by

```

Theorem 2 (parity). *For every $n, r \in \mathbb{N}$ with $n \bmod 5 = r$, $atc(n) \bmod 2 = pat2(r)$, where $pat2$ is the five-bit pattern of Listing 4. Equivalently, in the 1-based convention, $a(n)$ is odd if and only if $n \equiv 2$ or $3 \pmod{5}$.*

Theorem 3 (residue modulo 4). *For every $n, r \in \mathbb{N}$ with $n \bmod 10 = r$, $atc(n) \bmod 4 = pat4(r)$, where $pat4$ is the ten-bit pattern of Listing 5: 2, 3, 1, 2, 2, 0, 1, 1, 0, 0 along the residue classes $1, \dots, 10$ of the 1-based convention.*

Listing 4: The parity pattern `pat2` and the formal statement of Theorem 2, verbatim from the frozen statement file (lines 96–107, sha256 4cb10838).

```

1  /-- **余数样式 pat2** (mod 5 留数类  $\mapsto$  mod 2 余数, 0-based) :
2  0 $\mapsto$ 0, 1 $\mapsto$ 1, 2 $\mapsto$ 1, 其余 $\mapsto$ 0 (即  $\text{atc } n$  为奇  $\iff n \equiv 1$  或  $2 \pmod{5}$ );
3  1-based 口径:  $a(k)$  为奇  $\iff k \equiv 2$  或  $3 \pmod{5}$ )。 -/
4  def pat2 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
5    | 0 => 0
6    | 1 => 1
7    | 2 => 1
8    | _ => 0
9
10 /-- **T2 (伴随) **:  $\text{atc}$  的 mod 2 余数仅依赖  $n \bmod 5$ , 样式为 pat2。
    -/
11 theorem atc_mod2_period5 (n r :  $\mathbb{N}$ ) (hr : n % 5 = r) :
12   atc n % 2 = pat2 r := by

```

Listing 5: The mod-4 pattern `pat4` and the formal statement of Theorem 3, verbatim from the frozen statement file (lines 110–126, sha256 4cb10838).

```

1  /-- **余数样式 pat4** (mod 10 留数类  $\mapsto$  mod 4 余数, 0-based) :
2   $r \mapsto a(r+1) \bmod 4$ ,  $r = 0..9$  依次 (2,3,1,2,2,0,1,1,0,0)
3  (1-based 口径:  $a(k) \bmod 4$  按  $k \bmod 10$ ,  $k=1..10$  同为
4    (2,3,1,2,2,0,1,1,0,0))。 -/
5  def pat4 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
6    | 0 => 2
7    | 1 => 3
8    | 2 => 1
9    | 3 => 2
10   | 4 => 2
11   | 5 => 0
12   | 6 => 1
13   | 7 => 1
14   | _ => 0
15
16 /-- **T3 (伴随) **:  $\text{atc}$  的 mod 4 余数仅依赖  $n \bmod 10$ , 样式为 pat4。
    -/
17 theorem atc_mod4_period10 (n r :  $\mathbb{N}$ ) (hr : n % 10 = r) :
    atc n % 4 = pat4 r := by

```

All three statements are of the pipeline’s highest grade, complete proof: no sorry, axiom dependencies inside the whitelist, and statement byte-identical to the frozen text. The grade is quoted from the audit record `audit/final-comb09-4ntwocorner.txt` of the source task; it is not a judgement made in this note.

Conjecture 1 (combinatorial bridge, not proved here). For every $n \geq 1$, $\text{atc}(n-1)$ equals $a(n)$, the number of matchings of the graph G_n obtained from the $4 \times n$ grid $P_4 \square P_n$ by deleting the upper and the lower corner vertex of the last column (so the last column keeps its two middle vertices).

3 Proof

All three proofs are strong induction on the index with the seed window discharged by computation and the step closed by congruence algebra. Theorem 1 is an equation between recurrences and needs one auxiliary lemma, `atc_sq`, which exhibits an even-lag recurrence for the main sequence as an integer linear combination of ten shifted instances of the defining recurrence; its coefficient list is exactly the subsequence signature (2), which is the algebraic face of the $p(x)p(-x)$ structure noted in Section 2. Listing 6 gives that lemma in full and Listing 7 the complete proof of Theorem 1. Theorems 2 and 3 share one skeleton — drop the residue parameter into an inner statement, collapse the recurrence modulo 2 (respectively 4), split the residue by `mod_cases` and close each branch by `decide` — and are excerpted at the inductive step with the first residue branch; both carry a raised `set_option maxHeartbeats` at declaration level, because the per-branch arithmetic accumulates. The complete 420-line final artifact ships in this deposit under `proofs/`.

Listing 6: The auxiliary lemma `atc_sq` in full, verbatim excerpt of the final artifact `final/01-comb09-4ntwocorner-proved.lean` (lines 106–150, sha256 `f1aa7099`; line endings normalized from CRLF, no other byte changed). The lemma is the only declaration the final file adds to the frozen statements; it lives in the proof layer and is allowed as such by the audit.

```

1  /-- **辅助引理 atc_sq (平方递推, T1 的关键中间件) **: 主列 atc 自身
   满足
2  偶滞后 18 的递推 (= 子列递推  $s(y)$  在  $y=x^2$  处的拉回  $s(x^2) = -p(x)p(-x)$ ,
    $q(E)a(n) = 0$  由主递推的 10 个相邻实例线性组合给出, 无需归纳)。
3  系数证书 (h0..h9)  $\mapsto$  ---- (1,1,23,29,91,111,41,41,9,1) 经 python
   符号展开
4  与 a(1..50) 数值逐项核验 (r1 准备记录)。 -/
5
6  theorem atc_sq (n : N) :
7      atc (n + 18) = 163 * atc (n + 16) - 2641 * atc (n + 14) +
      12479 * atc (n + 12)
8      - 22577 * atc (n + 10) + 16705 * atc (n + 8) - 5331 * atc (n
      + 6)

```

```

9      + 769 * atc (n + 4) - 47 * atc (n + 2) + atc n := by
10 have e : ∀ m : ℕ, atc (m + 9)
11   = 9 * atc (m + 8) + 41 * atc (m + 7) - 41 * atc (m + 6) -
12     111 * atc (m + 5)
13     + 91 * atc (m + 4) + 29 * atc (m + 3) - 23 * atc (m + 2) -
14     atc (m + 1) + atc m :=
15   fun m => by simp only [atc]
16 have h0 : atc (n + 9) = 9 * atc (n + 8) + 41 * atc (n + 7) - 41
17   * atc (n + 6)
18   - 111 * atc (n + 5) + 91 * atc (n + 4) + 29 * atc (n + 3) -
19   23 * atc (n + 2)
20   - atc (n + 1) + atc n := e n
21 have h1 : atc (n + 10) = 9 * atc (n + 9) + 41 * atc (n + 8) - 41
22   * atc (n + 7)
23   - 111 * atc (n + 6) + 91 * atc (n + 5) + 29 * atc (n + 4) -
24   23 * atc (n + 3)
25   - atc (n + 2) + atc (n + 1) := e (n + 1)
26 have h2 : atc (n + 11) = 9 * atc (n + 10) + 41 * atc (n + 9) -
27   41 * atc (n + 8)
28   - 111 * atc (n + 7) + 91 * atc (n + 6) + 29 * atc (n + 5) -
29   23 * atc (n + 4)
30   - atc (n + 3) + atc (n + 2) := e (n + 2)
31 have h3 : atc (n + 12) = 9 * atc (n + 11) + 41 * atc (n + 10) -
32   41 * atc (n + 9)
33   - 111 * atc (n + 8) + 91 * atc (n + 7) + 29 * atc (n + 6) -
34   23 * atc (n + 5)
35   - atc (n + 4) + atc (n + 3) := e (n + 3)
36 have h4 : atc (n + 13) = 9 * atc (n + 12) + 41 * atc (n + 11) -
37   41 * atc (n + 10)
38   - 111 * atc (n + 9) + 91 * atc (n + 8) + 29 * atc (n + 7) -
39   23 * atc (n + 6)
40   - atc (n + 5) + atc (n + 4) := e (n + 4)
41 have h5 : atc (n + 14) = 9 * atc (n + 13) + 41 * atc (n + 12) -
42   41 * atc (n + 11)
43   - 111 * atc (n + 10) + 91 * atc (n + 9) + 29 * atc (n + 8) -
44   23 * atc (n + 7)
45   - atc (n + 6) + atc (n + 5) := e (n + 5)
46 have h6 : atc (n + 15) = 9 * atc (n + 14) + 41 * atc (n + 13) -
47   41 * atc (n + 12)
48   - 111 * atc (n + 11) + 91 * atc (n + 10) + 29 * atc (n + 9)
49   - 23 * atc (n + 8)
50   - atc (n + 7) + atc (n + 6) := e (n + 6)
51 have h7 : atc (n + 16) = 9 * atc (n + 15) + 41 * atc (n + 14) -
52   41 * atc (n + 13)
53   - 111 * atc (n + 12) + 91 * atc (n + 11) + 29 * atc (n + 10)

```

```

37   - 23 * atc (n + 9)
38   - atc (n + 8) + atc (n + 7) := e (n + 7)
39   have h8 : atc (n + 17) = 9 * atc (n + 16) + 41 * atc (n + 15) -
40     41 * atc (n + 14)
41     - 111 * atc (n + 13) + 91 * atc (n + 12) + 29 * atc (n + 11)
42     - 23 * atc (n + 10)
43     - atc (n + 9) + atc (n + 8) := e (n + 8)
44   have h9 : atc (n + 18) = 9 * atc (n + 17) + 41 * atc (n + 16) -
45     41 * atc (n + 15)
46     - 111 * atc (n + 14) + 91 * atc (n + 13) + 29 * atc (n + 12)
47     - 23 * atc (n + 11)
48     - atc (n + 10) + atc (n + 9) := e (n + 9)
49   linear_combination h9 + 9 * h8 - 41 * h7 - 41 * h6 + 111 * h5 +
50     91 * h4
51   - 29 * h3 - 23 * h2 + h1 + h0

```

Listing 7: Proof of Theorem 1 in full, verbatim excerpt of the final artifact (lines 155–194, sha256 f1aa7099; line endings normalized from CRLF).

```

1  theorem atc_interleave (k : ℕ) :
2    atc (2 * k) = ot k ∧ atc (2 * k + 1) = et k := by
3    induction k using Nat.strong_induction_on with
4    | h k ih =>
5      by_cases hk : k < 9
6      · interval_cases k <;> decide
7      · obtain ⟨j, rfl⟩ : ∃ j, k = j + 9 := ⟨k - 9, by omega⟩
8      constructor
9      · have e18 : 2 * (j + 9) = 2 * j + 18 := by ring
10       rw [e18, atc_sq]
11       simp only [ot]
12       rw [show 2 * j + 16 = 2 * (j + 8) by ring,
13         show 2 * j + 14 = 2 * (j + 7) by ring,
14         show 2 * j + 12 = 2 * (j + 6) by ring,
15         show 2 * j + 10 = 2 * (j + 5) by ring,
16         show 2 * j + 8 = 2 * (j + 4) by ring,
17         show 2 * j + 6 = 2 * (j + 3) by ring,
18         show 2 * j + 4 = 2 * (j + 2) by ring,
19         show 2 * j + 2 = 2 * (j + 1) by ring,
20         (ih (j + 8) (by omega)).1, (ih (j + 7) (by omega)).1,
21         (ih (j + 6) (by omega)).1, (ih (j + 5) (by omega)).1,
22         (ih (j + 4) (by omega)).1, (ih (j + 3) (by omega)).1,
23         (ih (j + 2) (by omega)).1, (ih (j + 1) (by omega)).1,
24         (ih j (by omega)).1]
25       · have e19 : 2 * (j + 9) + 1 = 2 * j + 1 + 18 := by ring
26       rw [e19, atc_sq]

```

```

27     simp only [et]
28     rw [show 2 * j + 1 + 16 = 2 * (j + 8) + 1 by ring,
29         show 2 * j + 1 + 14 = 2 * (j + 7) + 1 by ring,
30         show 2 * j + 1 + 12 = 2 * (j + 6) + 1 by ring,
31         show 2 * j + 1 + 10 = 2 * (j + 5) + 1 by ring,
32         show 2 * j + 1 + 8 = 2 * (j + 4) + 1 by ring,
33         show 2 * j + 1 + 6 = 2 * (j + 3) + 1 by ring,
34         show 2 * j + 1 + 4 = 2 * (j + 2) + 1 by ring,
35         show 2 * j + 1 + 2 = 2 * (j + 1) + 1 by ring,
36         (ih (j + 8) (by omega)).2, (ih (j + 7) (by omega)).2,
37         (ih (j + 6) (by omega)).2, (ih (j + 5) (by omega)).2,
38         (ih (j + 4) (by omega)).2, (ih (j + 3) (by omega)).2,
39         (ih (j + 2) (by omega)).2, (ih (j + 1) (by omega)).2,
40         (ih j (by omega)).2]

```

Listing 8: Theorem 2: the clean inner lemma, the induction shell, the mod-2 collapse and the first residue branch, verbatim excerpt of the final artifact (lines 200–228, sha256 f1aa7099; line endings normalized from CRLF). The theorem closes at line 277 of the artifact by rewriting the hypothesis into the inner statement; the other four residue branches are identical in form with different residues.

```

1  have main : ∀ n : ℕ, atc n % 2 = pat2 (n % 5) := by
2    intro n
3    induction n using Nat.strong_induction_on with
4    | h n ih =>
5      by_cases hn : n < 9
6      · interval_cases n <=> decide
7      · obtain ⟨j, rfl⟩ : ∃ j, n = j + 9 := ⟨n - 9, by omega⟩
8        have h1 : atc (j + 9) % 2
9          = (atc (j + 8) % 2 + atc (j + 7) % 2 - atc (j + 6) % 2
10            - atc (j + 5) % 2
11              + atc (j + 4) % 2 + atc (j + 3) % 2 - atc (j + 2)
12                % 2 - atc (j + 1) % 2
13                + atc j % 2) % 2 := by
14          simp only [atc]
15          omega
16          rw [h1, ih (j + 8) (by omega), ih (j + 7) (by omega), ih (
17            j + 6) (by omega),
18              ih (j + 5) (by omega), ih (j + 4) (by omega), ih (j +
19            3) (by omega),
20              ih (j + 2) (by omega), ih (j + 1) (by omega), ih j (by
21            omega)]
22          mod_cases hj : j % 5

```

```

18 · have hj0 : j % 5 = 0 := by simp [Nat.ModEq] using hj
19   have hj1 : (j + 1) % 5 = 1 := by omega
20   have hj2 : (j + 2) % 5 = 2 := by omega
21   have hj3 : (j + 3) % 5 = 3 := by omega
22   have hj4 : (j + 4) % 5 = 4 := by omega
23   have hj5 : (j + 5) % 5 = 0 := by omega
24   have hj6 : (j + 6) % 5 = 1 := by omega
25   have hj7 : (j + 7) % 5 = 2 := by omega
26   have hj8 : (j + 8) % 5 = 3 := by omega
27   have hj9 : (j + 9) % 5 = 4 := by omega
28   rw [hj0, hj1, hj2, hj3, hj4, hj5, hj6, hj7, hj8, hj9]
29   decide

```

Listing 9: Theorem 3: the same skeleton at period 10 — the mod-4 collapse and the first residue branch, verbatim excerpt of the final artifact (lines 283–311, sha256 f1aa7099; line endings normalized from CRLF).

```

1  have main : ∀ n : ℕ, atc n % 4 = pat4 (n % 10) := by
2    intro n
3    induction n using Nat.strong_induction_on with
4      | h n ih =>
5        by_cases hn : n < 9
6        · interval_cases n <=> decide
7        · obtain ⟨j, rfl⟩ : ∃ j, n = j + 9 := ⟨n - 9, by omega⟩
8          have h1 : atc (j + 9) % 4
9            = (atc (j + 8) % 4 + atc (j + 7) % 4 - atc (j + 6) % 4
10              + atc (j + 5) % 4
11                - atc (j + 4) % 4 + atc (j + 3) % 4 + atc (j + 2)
12                  % 4 - atc (j + 1) % 4
13                    + atc j % 4) % 4 := by
14            simp only [atc]
15            omega
16            rw [h1, ih (j + 8) (by omega), ih (j + 7) (by omega), ih (
17              j + 6) (by omega),
18              ih (j + 5) (by omega), ih (j + 4) (by omega), ih (j +
19              3) (by omega),
20              ih (j + 2) (by omega), ih (j + 1) (by omega), ih j (by
21              omega)]
22          mod_cases hj : j % 10
23          · have hj0 : j % 10 = 0 := by simp [Nat.ModEq] using hj
24            have hj1 : (j + 1) % 10 = 1 := by omega
25            have hj2 : (j + 2) % 10 = 2 := by omega
26            have hj3 : (j + 3) % 10 = 3 := by omega
27            have hj4 : (j + 4) % 10 = 4 := by omega
28            have hj5 : (j + 5) % 10 = 5 := by omega

```

```

24      have hj6 : (j + 6) % 10 = 6 := by omega
25      have hj7 : (j + 7) % 10 = 7 := by omega
26      have hj8 : (j + 8) % 10 = 8 := by omega
27      have hj9 : (j + 9) % 10 = 9 := by omega
28      rw [hj0, hj1, hj2, hj3, hj4, hj5, hj6, hj7, hj8, hj9]
29      decide

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (leanprover/lean4:v4.34.0), mathlib4 v4.34.0, revision 5ed29652 [1, 2, 3]; this note is typeset with tectonic 0.17.0.
- Statement integrity: the frozen statement file `formalized/01-comb09-4ntwocorner-statements.lean` (127 lines) has sha256 4cb1083846bcc64e249717309e63d071d0ecf5bb03f23d85affeaab8152ce005; the final file `final/01-comb09-4ntwocorner-proved.lean` (420 lines, CRLF at rest) has sha256 f1aa7099697fe5706bc5f4dd7124d06f4f92d6c2273b349b862c4b963806f53f. The final file was compared with the frozen file by symmetric difference over the declaration region: 8 of 8 frozen declarations byte-identical, all 8 docstrings contained verbatim; the only addition is the proof-layer lemma `atc_sq`.
- Kernel check: compiles with 0 `sorry` (whole-file scan including comments, case-insensitive, also for `admit/sorryAx`); the strict re-compile exits 0 with zero diagnostics. The axiom audit output, verbatim (kept literal; the smaller monospace width is typesetting only):


```

1 AXIOMS atc_interleave [propext, Classical.choice, Quot.sound]
2 AXIOMS atc_mod2_period5 [propext, Quot.sound]
3 AXIOMS atc_mod4_period10 [propext, Quot.sound]

```
- Grading by the audit department: all three theorems *complete proof*; lowest grade reached for the case = *complete proof*; no scaffolding route was used.
- Audit chain (originals in this deposit under `audit/`): final adjudication `final-comb09-4ntwocorner.txt` (four independent checks), the axiom printouts `axioms-atc_interleave.txt`, `axioms-atc_mod2_period5.txt` and `axioms-atc_mod4_period10.txt`,

the rechecker’s independent re-run `axioms-recheck.txt`, this note’s lint and compile logs and listing byte-exactness record, and the full C9 search log `c9-record.md`. The semantic-fidelity check (roundtrip re-translation) of the underlying task ran on the same model node as its formalisation; that independence weakness is mitigated, not removed, by kernel adjudication and manual review.

5 Novelty evidence

Adjudication copied from the pipeline’s exclusivity review (C9 gate), which ran before the problem was accepted and was repeated in a third, deepest round by workers who re-derived the data independently rather than trusting the external report; all three rounds returned *no occupying record*, and the third round graded the value tier strictly as *new sequence*, correcting the earlier “new sequence and new recurrence” wording. It is not a claim of new mathematics and not a first-discovery claim. Summary by channel, from the third-round record (`tasks/20260928-comb0709-deeprecheck/recheck/deep-recheck-combm-03.md`, kept in the source repository); the full query log, including every zero-hit query, is in this deposit at `audit/c9-record.md`.

observed	queries found 0:	result
OEIS sequence search [6]	0 new strings	new lists (the main sequence accumulated 0); independent query strings across rounds: head, transition, proposed, new-proposed variants, characteristic polynomial and numerator strings, next-0 subsequences, tail windows, difference windows)
OEIS pre-emptive subsequences	included above	are like search-0 results empty up to round 0;
ch14111 / n31e / Math StackExchange / Crouzet	20 new alpha groups	new name-shaped lists, two n31e problem entries total=0
Boomerangs, almost neighbours [7]	full-text read	There are 4 so-called general lists with a differently-shaped statistic; the sequence does not occur
OEIS record deep read	ALLIDENT, ALLIDROW full columns	neither locally literature comes only unchanged grade, no defect variant criterion
Shower base	available, complete, unapplied	are new-shaped objects available has the matching problem [1] but no counting theorem
public form accuracy	only search as newly computed large terms	total=0

The review’s machine checks, re-derived independently in round 3: the 12-term head was reproduced term by term by three heterogeneous methods; the order-9 recurrence holds on 50 computed terms; the minimal order over $\mathbb{Q}$ is exactly 9 (no order-8 fit). Two occupancies were established and are carried as boundaries: A033507 [9], whose recurrence line is the namesake recurrence of this note, and A129113 [10] for the perfect-matching subcolumn.

The C9 verdict of the third round is reproduced verbatim below (the full record ships as `audit/c9-record.md` in this deposit):

- 1 ****C9 四态：查无占位（三轮一致，本审支持前两审结论）。****
- 2 ****价值分级：新序列****（九阶递推与已挂名的 **A033507** 完全同谱——前两审已注，本审机器确认：「新序列-新递推」措辞本审维持修正为「新序列」）。

The occupancy fences of the source claim log (`tasks/20260928-comb09-4ntwocorner/phase0/claims.md` §0, kept in the source repository) bound every wording of this note and are reproduced verbatim below:

¹ **占位围栏：价值级仅「新序列」——九阶递推即 A033507（无缺陷 $4 \times n$ 全匹配）的挂名递推（同谱姊妹序列），禁止对递推本身作新颖性主张；PM 子列 = OEIS A129113 已挂名占位，禁止以其作新颖性主张。**

The residual gaps of the review are disclosed with it: the general web search layer was unreachable in the review environment, so literature coverage is carried by four structured channels (OpenAlex, zbMATH, arXiv, Crossref); the Math StackExchange native API was throttled at review time; the mother-family full texts (Lundow 1996/1998, Read 1982) and Guichard 2008 were not read in full (the latter only possibly touches the already-occupied perfect-matching face); and the machine verification of the recurrence stops at 50 computed terms. If any of these channels later returns a hit on this sequence, the tier above weakens accordingly, and this deposit will be amended by a later version rather than by editing this record.

6 Scope and limitations

Grade and boundary, stated as the audit states them. (a) The theorems cover the recurrence-defined sequences only; their equality with matching counts of the notched grid is not kernel-proved and is Conjecture 1. Its empirical support is enumeration by two independently written programs agreeing on their overlap; the enumerators themselves are not verified programs, and the scripts and their outputs stay in the working repository under `tasks/20260928-comb09-4ntwocorner/`. (b) Priority boundary: the order-9 recurrence is the namesake recurrence of OEIS A033507 [9] — the undamaged $4 \times n$ grid, a same-spectrum sister sequence — so nothing is claimed about the recurrence itself; the perfect-matching subcolumn is OEIS A129113 [10], already occupied, and is claimed by nobody here. The value tier of this note is strictly “new sequence” (initial values) plus the three structural theorems, and the novelty evidence is negative search evidence with disclosed holes. (c) Indexing is 0-based in the formal layer ($\mathbf{atc}(n) = a(n+1)$, $\mathbf{ot}(k) = a(2k+1)$, $\mathbf{et}(k) = a(2k+2)$); the 1-based reading is used for the combinatorial statements. (d) The recurrences are computational findings: exact rational minimal-order fits (order exactly 9; no order-8 fit) verified against enumeration, not derived from a transfer-matrix argument; the theorems assert the periods 5 and 10 and the pattern tables, not their minimality, which remains a computed fact. The mod-3 period 364 was measured the same way and was deliberately not formalised in this batch. (e) The gate-four sign-off of the underlying task was exercised by the author’s instruction to deposit this claim on 2026-09-28, while the task report file still

carries its draft header (“pending manual sign-off at gate four”); both facts are recorded here as they stand, and the task’s own artifacts are not edited. (f) The roundtrip semantic check of the underlying task ran on the same model node as its formalisation; the independence risk is mitigated, not removed, by kernel adjudication and manual review. (g) This note carries no literature review and is a deposit record, not a survey; the narrative paper for the same result is separate.

Data availability

The deposit package for this claim contains the claim note (LaTeX source and PDF), the Lean sources (`proofs/`: frozen statement file, final proof file, `lean-toolchain`), the verification and audit logs (final adjudication, axiom printouts and recheck, this note’s lint and compile logs, listing byte-exactness record), and the full C9 query log with zero-hit entries (`audit/`). Zenodo: the DOI is an unfilled field, `RESERVED-DOI`; it is reserved by the author at upload and recorded in `metadata/README.md` of this package once created; the upload steps are listed in `UPLOAD.md` in the source repository, not shipped with this deposit. Public mirror of this deposit: the claim lane’s repository <https://github.com/Aurora0134/ai4math-claims>, directory `grid4n-col-2notch/` (pushed under the author’s instruction of 2026-09-28; the commit and timestamp are recorded in the source repository’s `claims/grid4n-col-2notch/card.md`). Claim note under CC BY 4.0, code under MIT.

Use of generative AI

(a) AI performed: candidate generation and the exclusivity review, autoformalisation of the five definitions and three statements, the roundtrip semantic check, proof search and proof-script writing, error classification and repair, the assembly of the final file, the audit re-runs, the prose translation of the proofs, and the assembly of this note including its verbatim listings. (b) Model, node and budget: all task-stage sampling ran on one node, wire-id `anthropic/a6api-main/ki4i-k3`, with no cross-node switching, and consumed 9 worker dispatches of a permitted 20, 1 LLM call of a permitted 6 and 25 kernel compiles of a permitted 32, copied verbatim from the task ledger `budget.log` (a source repository file, not shipped with this deposit). This note consumed 0 pipeline LLM calls; its LaTeX compilations are counted in this claim lane’s own ledger at the same source

repository location, not shipped with this deposit. The node record for 2026-09-28 carries a drift observation — the persistent relay record shows both `anthropic/a6api-main/kimi-k3` and `stepfun/step-5-preview` on that date, and the two independent re-audit dispatches of the paper lane ran on the latter; this is disclosed rather than smoothed over. (c) Final adjudication: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and the axiom audit reports only `{propext, Quot.sound, Classical.choice}` or a subset thereof for each of `atc_interleave`, `atc_mod2_period5` and `atc_mod4_period10`. (d) The AI system is not an author; the human author reviewed the evidence and is responsible for all content.

Author contributions

CRediT-style: the human author adjudicated topic selection, statement semantics and the deposit decision; the AI4Math pipeline (an LLM) generated the candidate propositions, the proof searches and the text drafts. The AI system is not listed as an author.

References

- [1] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. *Automated Deduction — CADE-28*, LNCS, pp. 625–635, 2021. https://doi.org/10.1007/978-3-030-79876-5_37
- [2] The mathlib Community. The Lean mathematical library. *CPP 2020: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 367–381, 2020. <https://doi.org/10.1145/3372885.3373824>
- [3] The mathlib Community. The mathlib4 repository, tag v4.34.0 (commit 5ed2965256430c3649e86755f9576b54eca72435); its `lean-toolchain` pin reads `leanprover/lean4:v4.34.0`. <https://github.com/leanprover-community/mathlib4/tree/v4.34.0>
- [4] O. J. Heilmann and E. H. Lieb. Theory of monomer-dimer systems. *Communications in Mathematical Physics* 25 (1972), pp. 190–232. <https://doi.org/10.1007/BF01877590>

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- [8] The OEIS Foundation. *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/> — searched for the sequence (1)’s initial values, its subsequences and variants, with no entry found (see Section 5).
- [9] The OEIS Foundation. Entry A033507 (P. H. Lundow): number of matchings in graph $P_4 \times P_n$; recurrence line $a(n) = 9a(n-1) + 41a(n-2) - 41a(n-3) - 111a(n-4) + 91a(n-5) + 29a(n-6) - 23a(n-7) - a(n-8) + a(n-9)$ and index-line signature (9, 41, -41, -111, 91, 29, -23, -1, 1) as displayed. *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A033507>
- [10] The OEIS Foundation. Entry A129113 (R. L. Bagula, 2007): expansion of $x^3/(1 - x - 5x^2 - x^3 + x^4)$; data 0, 0, 0, 1, 1, 6, 12, 42, 107, 323, 888, 2568, 7224, 20629, 58429, ... as displayed. *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A129113>
- [11] The mathlib Community. `Matching.lean`, in `Mathlib/Combinatorics/SimpleGraph/`, `mathlib4`, tag `v4.34.0`: defines the predicates `IsMatching` and `IsPerfectMatching` and structural lemmas, and contains no matching-count function. <https://github.com/leanprover-community/mathlib4/blob/v4.34.0/Mathlib/Combinatorics/SimpleGraph/Matching.lean>